



exercise2.py x

```
1 class Airplane:
2     def __init__(self, first_class, business_class, coach):
3         self.first_class, self.business_class, self.coach = first_class, business_class, coach
4     def total(self):
5         return self.first_class + self.business_class + self.coach
6
7 class Train:
8     def __init__(self, car1, car2, car3, car4, car5):
9         self.car1, self.car2, self.car3, self.car4, self.car5 = car1, car2, car3, car4, car5
10    def total(self):
11        return self.car1 + self.car2 + self.car3 + self.car4 + self.car5
12
13 def passengers(obj):
14     print(f'There are {obj.total()} passengers on board.')
15
16 passengers(Airplane(20, 10, 5))
17 passengers(Train(20, 10, 5, 1, 2))
```

Guide x

Collapse

Polymorphism -> Coding Exercises - Polymor...



Polymorphism Exercise 2

Exercise 2

Complete the `Airplane` and `Train` classes so that when an instance of each is passed to the `passengers` function, they will return the total number of passengers on board.

TRY IT

There are 35 passengers on board.
There are 38 passengers on board.

Submit your code for evaluation

Here is one possible solution:

```
class Airplane:
    def __init__(self, first_class, business_class, coach):
        self.first_class = first_class
        self.business_class = business_class
        self.coach = coach

    def total(self):
        return self.first_class + self.business_class + self.coach

class Train:
    def __init__(self, car1, car2, car3, car4, car5):
        self.car1 = car1
        self.car2 = car2
        self.car3 = car3
        self.car4 = car4
        self.car5 = car5

    def total(self):
        return self.car1 + self.car2 + self.car3 + self.car4 + self.car5

def passengers(obj):
    print(f'There are {obj.total()} passengers on board.')
```

Check It!

LAST RUN on 12/28/2023,
2:31:08 AM

Test passed

Next